

LAB 1 - FACIAL EMOTION RECOGNITION

Deep Learning on FER-2013 with Real-time DroidCam Inference

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1. Objective and Dataset

The objective is to build and compare two facial emotion recognition models, evaluate them on FER-2013, and deploy a real-time pipeline that receives video from a phone camera, detects faces, classifies emotions, and displays bounding boxes, confidence scores, and FPS.

- Dataset: FER-2013, 35,887 grayscale facial images at 48 x 48 pixels.
- Classes: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral.
- Official split used: Training = 28,709; PublicTest = 3,589; PrivateTest = 3,589.
- Class imbalance is substantial, especially for Disgust; class weighting is therefore applied during training.

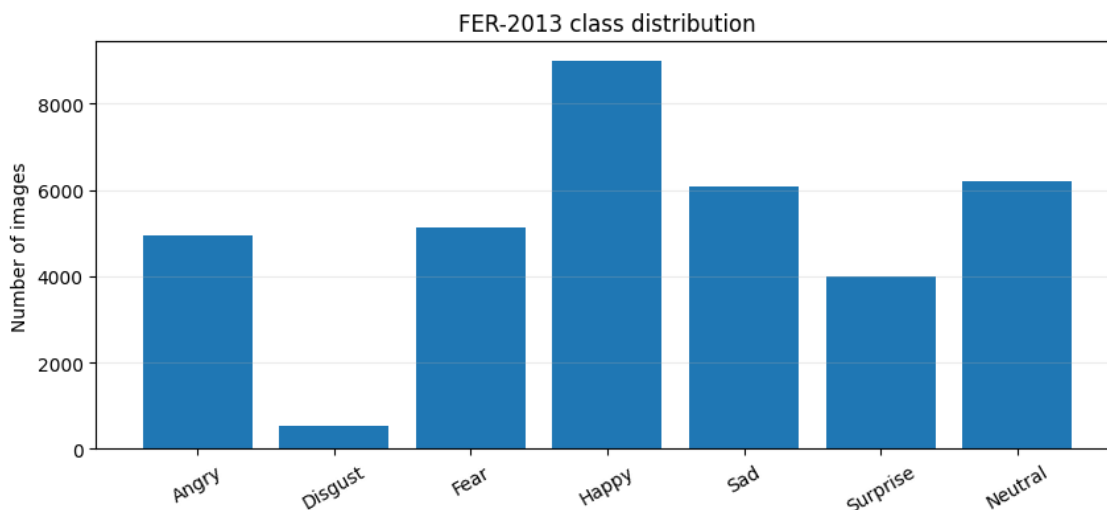


Figure 1. FER-2013 training-set class distribution.

2. Preprocessing and Evaluation Protocol

M1 uses the original grayscale representation, resizing/cropping each face to 48 x 48, converting to float, and normalizing with $(x - 0.5) / 0.5$. M2 converts grayscale faces to three channels, resizes them to 224 x 224, and applies ImageNet normalization to match the pretrained MobileNetV3-Small backbone.

- Training augmentation is intentionally mild: horizontal flip, small rotation, translation, and scale changes.
- M1 uses softened square-root inverse-frequency class weights and label smoothing (0.02).

- Model selection uses validation performance only; PrivateTest is evaluated after model selection.
- Main metrics: test accuracy, macro F1, confusion matrix, parameter count, and CPU inference speed.

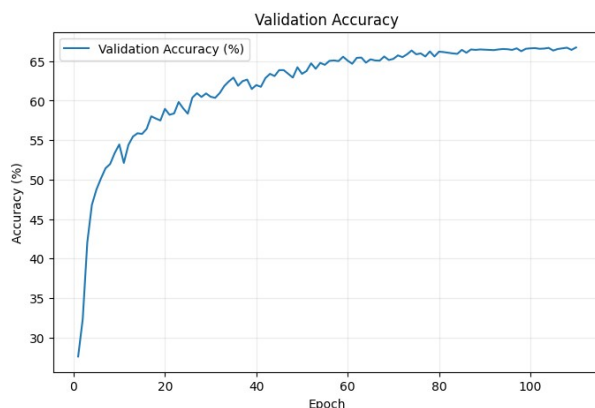
3. Model M1 - FERAttentionLite (Custom CNN)

FERAttentionLite is a custom lightweight CNN trained from scratch. It uses four convolutional stages with Batch Normalization, ReLU, Max Pooling and Dropout. The third stage contains both channel attention and spatial attention, followed by global average pooling and a compact classifier.

Stage	Main layers	Purpose
Block 1	Conv 1→32→32 + Pool + Dropout	Low-level edges/textures
Block 2	Conv 32→64→64 + Pool + Dropout	Mid-level facial patterns
Block 3	Conv 64→128→128 + Channel + Spatial Attention	Emphasize useful feature channels/regions
Block 4	Conv 128→256→256 + Dropout	High-level expression features
Head	AdaptiveAvgPool + FC 256→128→7	Compact 7-class classification

3.1 Training configuration and result

- Optimizer: AdamW; initial learning rate $3e-4$; weight decay $1e-4$.
- Scheduler: cosine annealing. Batch size: 512. Training used AMP.
- Best checkpoint: epoch 94, selected by validation macro F1.
- Validation accuracy: 66.54%; validation macro F1: 0.6480.
- PrivateTest accuracy: 68.54%; PrivateTest macro F1: 0.6690.
- Trainable parameters: 1,211,033.



FERAttentionLite — PrivateTest Confusion Matrix

	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral
Angry	306	15	41	13	74	2	40
Disgust	6	43	1	1	2	1	1
Fear	75	6	195	12	128	53	59
Happy	20	0	10	767	31	23	28
Sad	49	1	31	21	377	6	109
Surprise	9	1	38	21	4	333	10
Neutral	26	3	26	30	94	8	439
	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral

True label

Predicted label

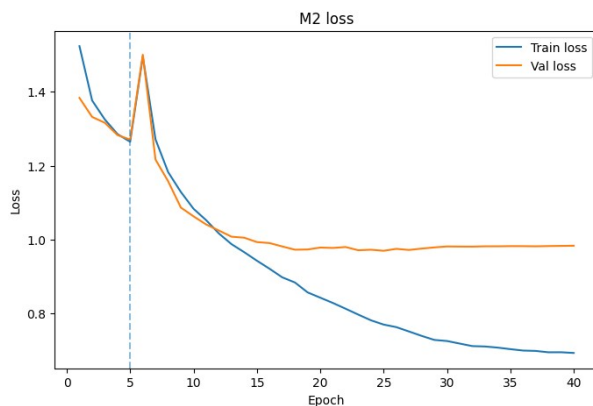
Figure 2. M1 validation accuracy curve (left) and PrivateTest confusion matrix (right).

The learning curve increases steadily and stabilizes around the mid-60% validation range, indicating convergence without severe instability. The confusion matrix shows that Fear, Sad, and Neutral are among the most difficult classes to separate. The largest M1 confusion pairs include Fear→Sad (128), Sad→Neutral (109), Neutral→Sad (94), Fear→Angry (75), and Angry→Sad (74).

4. Model M2 - MobileNetV3-Small Transfer Learning

M2 uses MobileNetV3-Small initialized with ImageNet weights. The original classifier is replaced by a seven-class output layer. Training is performed in two stages: first training the classifier head with the feature extractor frozen, then unfreezing the network and fine-tuning it with smaller learning rates.

- Input: 224 x 224 RGB generated from FER grayscale images; ImageNet mean/std normalization.
- Stage 1: frozen feature extractor; only classifier head is trained.
- Stage 2: full-network fine-tuning with lower backbone learning rate.
- Best checkpoint: epoch 33 (fine-tuning stage).
- Validation accuracy: 65.67%; validation macro F1: 0.6433.
- PrivateTest accuracy: 67.54%; PrivateTest macro F1: 0.6591.
- Trainable parameters: 1,525,031.



M2 MobileNetV3-Small — Confusion Matrix

	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral
Angry	280	7	55	16	66	13	54
Disgust	14	36	0	1	1	1	2
Fear	60	4	263	20	95	38	48
Happy	10	1	19	775	21	16	37
Sad	61	2	68	21	340	6	96
Surprise	15	1	36	22	10	312	20
Neutral	33	3	31	30	107	4	418
	Angry	Disgust	Fear	Happy	Sad	Surprise	Neutral

Figure 3. M2 validation accuracy curve (left) and PrivateTest confusion matrix (right).

4.1 Interpretation

Although transfer learning provides a strong initialization, M2 does not exceed M1 on this task. FER-2013 contains tiny 48 x 48 grayscale faces, while ImageNet pretraining is based on larger natural RGB images. Resizing FER images to 224 x 224 and duplicating grayscale information across RGB channels introduces a

domain mismatch. The task-specific M1 architecture can therefore be slightly better despite having fewer parameters.

During live testing, M2 also showed more frame-to-frame label changes than M1. This does not invalidate the classifier, but it suggests lower temporal stability under small changes in pose, lighting, or detected face crop. The submitted version keeps the model output unsmoothed so that evaluation remains directly tied to the trained classifier.

5. Model Comparison and Error Analysis

Model	Test Acc.	Macro F1	Params	Latency	Predictor FPS
M1					
FERAttentionLite	68.54%	0.6690	1,211,033	4.01 ms	249.2
M2					
MobileNetV3-Small	67.54%	0.6591	1,525,031	11.65 ms	85.8

The local CPU benchmark was executed with the same benchmark script and 100 iterations for both models. The reported predictor FPS includes model-specific preprocessing and inference, but not camera capture, face detection, drawing, or GUI display. Therefore, predictor FPS must not be interpreted as full-pipeline FPS.

- M1 is +1.00 percentage point higher in test accuracy and +0.0099 higher in macro F1.
- M1 uses about 314k fewer parameters and is approximately 2.9x faster in the local predictor benchmark.
- Both models exceed their assignment thresholds: M1 $\geq 60\%$ and M2 $\geq 65\%$ test accuracy.
- Macro F1 is reported because FER-2013 is imbalanced and accuracy alone may hide poor performance on minority classes.

5.1 Misclassification analysis



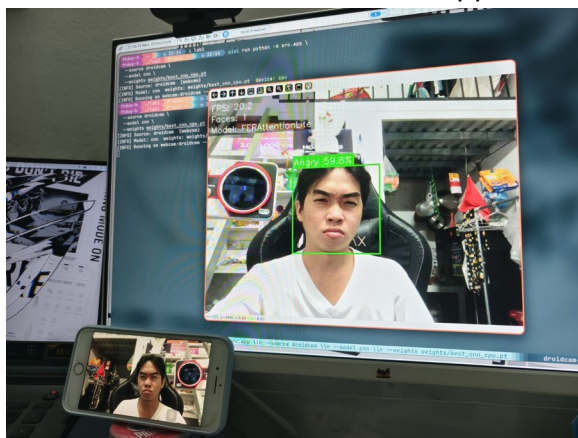
Figure 4. Representative misclassified PrivateTest examples: M1 (left) and M2 (right).

Most errors occur between visually similar or ambiguous expressions, especially Fear, Sad, Neutral, and Angry. Contributing factors include low 48 x 48 resolution, subtle facial muscle differences, head pose, illumination, partial occlusion, and subjective FER-2013 labels. Happy and Surprise are generally easier because they contain more distinctive mouth/eye patterns.

6. Real-time Phone-camera Deployment

The real-time application receives a DroidCam stream from a phone, detects faces with OpenCV, crops each face, applies the model-specific preprocessing, performs CPU inference, and overlays the predicted emotion, confidence, number of faces, model name, and FPS. The camera source is provided through the command line/configuration rather than hard-coded.

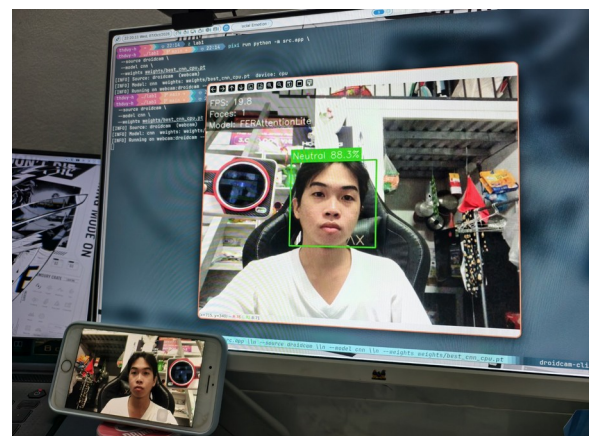
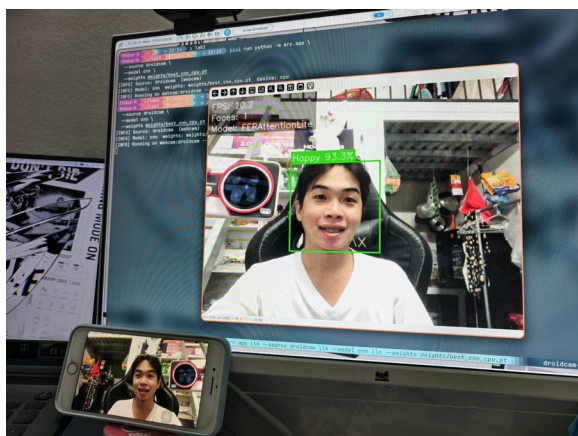
- Multiple detected faces are supported through batched predictor inference.
- M1 and M2 use separate predictor adapters, ensuring that each model receives the same preprocessing used during training.
- The pipeline supports webcam/IP-camera-style sources and handles real-time visualization.
- Both models achieved more than 10 FPS in full-pipeline testing. The M1 DroidCam demo below shows approximately 19.7-20.2 FPS.



Angry - 59.8%



Surprise - 98.0%



Happy - 93.3%**Neutral - 88.3%**

Figure 5. Real-time FERAttentionLite inference from DroidCam. The phone acts as the camera source; the laptop performs face detection and CPU inference.

7. Conclusion

The project successfully implements the complete facial emotion recognition pipeline required by the lab. FERAttentionLite achieved the best final result with 68.54% test accuracy and 0.6690 macro F1 while remaining smaller and faster than the transfer-learning model. MobileNetV3-Small achieved 67.54% accuracy and 0.6591 macro F1, satisfying the transfer-learning requirement. Both models operate above the required 10 FPS in real-time testing. Overall, the custom model provides the better accuracy-efficiency trade-off for this FER-2013 deployment.